Configuring data backups

Snap Creator Framework

NetApp
August 04, 2021

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Configuring data backups

After you install the required software components, follow these steps to complete the configuration:

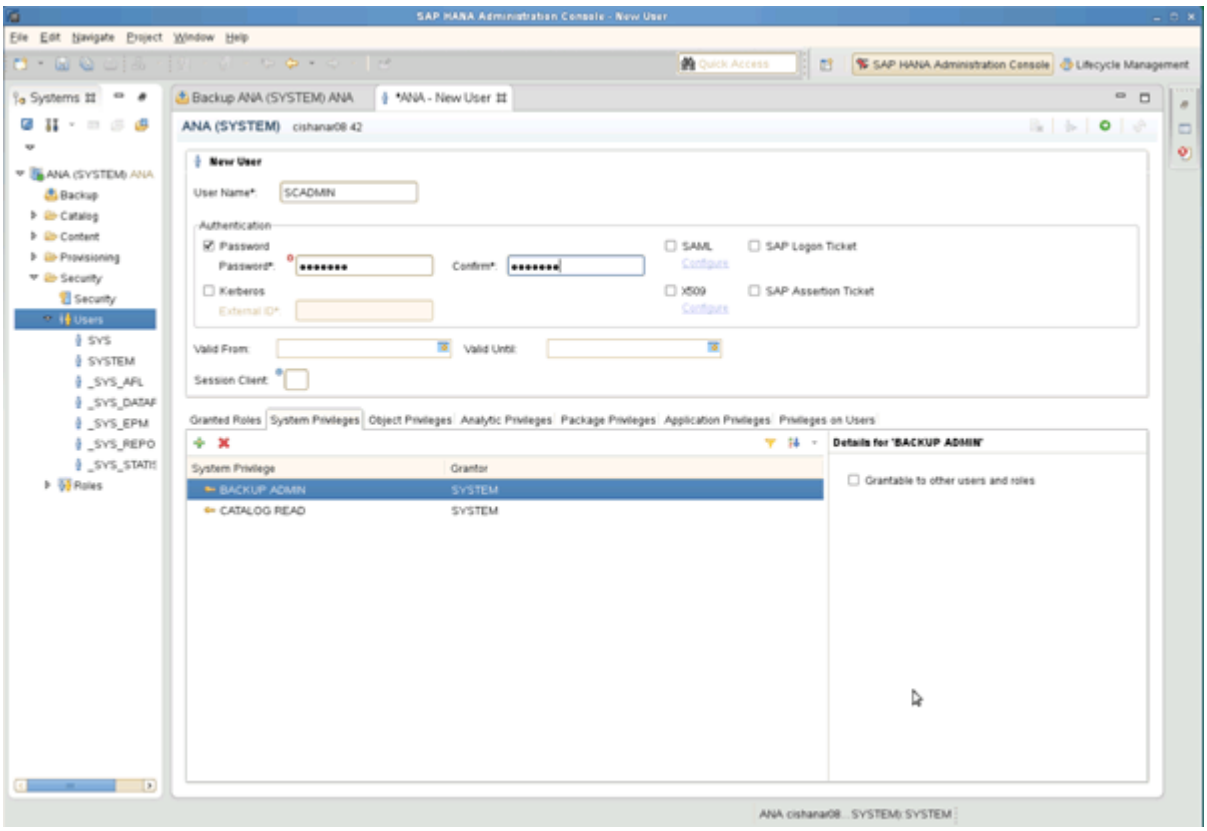
1. Configure a dedicated database user and the SAP HANA userstore.
2. Prepare SnapVault replication on all storage controllers.
3. Create volumes at secondary storage controller.
4. Initialize the SnapVault relationships for database volumes.
5. Configure Snap Creator.

Configuring the backup user and hdbuserstore

You should configure a dedicated database user within the HANA database to run the backup operations with Snap Creator. In a second step, you should configure an SAP HANA userstore key for this backup user. This userstore key is used within the configuration of the Snap Creator SAP HANA plug-in.

The backup user must have the following privileges:

- BACKUP ADMIN
- CATALOG READ



1. At the administration host, the host where Snap Creator got installed, a userstore key is configured for all database hosts that belong to the SAP HANA database. The userstore key is configured with the OS root user: hdbuserstore set keyhost 3[instance]15 userpassword

2. Configure a key for all four database nodes.

```
mgmtsrv01:/usr/sap/hdbclient32 # ./hdbuserstore set SCADMIN08
cishanar08:34215 SCADMIN Password
mgmtsrv01:/usr/sap/hdbclient32 # ./hdbuserstore set SCADMIN09
cishanar09:34215 SCADMIN Password
mgmtsrv01:/usr/sap/hdbclient32 # ./hdbuserstore set SCADMIN10
cishanar10:34215 SCADMIN password
mgmtsrv01:/usr/sap/hdbclient32 # ./hdbuserstore set SCADMIN11
cishanar11:34215 SCADMIN Password
mgmtsrv01:/usr/sap/hdbclient32 # ./hdbuserstore LIST
DATA FILE       : /root/.hdb/mgmtsrv01/SSFS_HDB.DAT

KEY SCADMIN08
  ENV : cishanar08:34215
  USER: SCADMIN
KEY SCADMIN09
  ENV : cishanar09:34215
  USER: SCADMIN
KEY SCADMIN10
  ENV : cishanar10:34215
  USER: SCADMIN
KEY SCADMIN11
  ENV : cishanar11:34215
  USER: SCADMIN
mgmtsrv01:/usr/sap/hdbclient32
```

Configuring SnapVault relationships

When you configure SnapVault relationships, the primary storage controllers must have a valid SnapRestore and SnapVault license installed. The secondary storage must have a valid SnapVault license installed.

1. Enable SnapVault and NDMP on the primary and the secondary storage controllers.

```
hana1a> options snapvault.enable on
hana1a> ndmp on
hana1a>
hana1b> options snapvault.enable on
hana1b> ndmpd on
hana1b
hana2b> options snapvault.enable on
hana2b> ndmpd on
hana2b>
```

2. On all primary storage controllers, configure the access to the secondary storage controller.

```
hana1a> options snapvault.access host=hana2b
hana1a>
hana1b> options snapvault.access host=hana2b
hana1b>
```

Using a dedicated network for replication traffic is recommended. In such cases, the host name of this interface at the secondary storage controller needs to be configured. Instead of hana2b, the host name could be hana2b-rep.

3. On the secondary storage controller, configure the access for all primary storage controllers.

```
hana2b> options snapvault.access host=hana1a,hana1b
hana2b>
```

Using a dedicated network for replication traffic is recommended. In such cases, the host name of this interface at the primary storage controllers needs to be configured. Instead of hana1b and hana1a the host name could be hana1a-rep and hana1b-rep.

Starting the SnapVault relationships

You need to start the SnapVault relationship with Data ONTAP operating in 7-Mode and clustered Data ONTAP.

Starting the SnapVault relationships with Data ONTAP operating in 7-Mode

You can start a SnapVault relationship with commands executed on the secondary storage system.

1. For storage systems running Data ONTAP operating in 7-Mode, you start the SnapVault relationships by running the following command:

```
hana2b> snapvault start -S hana1a:/vol/data_00001/mnt00001
/vol/backup_data_00001/mnt00001
Snapvault configuration for the qtree has been set.
Transfer started.
Monitor progress with 'snapvault status' or the snapmirror log.
hana2b>
hana2b> snapvault start -S hana1a:/vol/data_00003/mnt00003
/vol/backup_data_00003/mnt00003
Snapvault configuration for the qtree has been set.
Transfer started.
Monitor progress with 'snapvault status' or the snapmirror log.
hana2b>
hana2b> snapvault start -S hana1b:/vol/data_00002/mnt00002
/vol/backup_data_00002/mnt00002
Snapvault configuration for the qtree has been set.
Transfer started.
Monitor progress with 'snapvault status' or the snapmirror log.
hana2b>
```

It is recommended that you use a dedicated network for replication traffic. In that case, configure the host name of this interface at the primary storage controllers. Instead of hana1b and hana1a, the host name could be hana1a-rep and hana1b-rep.

Starting the SnapVault relationships with clustered Data ONTAP

You need to define a SnapMirror policy before you start a SnapVault relationship.

1. For storage systems running clustered Data ONTAP, you start the SnapVault relationships by running the following command.

```
hana::> snapmirror policy create -vserver hana2b -policy SV_HANA
hana::> snapmirror policy add-rule -vserver hana2b -policy SV_HANA
-snapmirror-label daily -keep 20
hana::> snapmirror policy add-rule -vserver hana2b -policy SV_HANA
-snapmirror-label hourly -keep 10

hana::> snapmirror policy show -vserver hana2b -policy SV_HANA

                   Vserver: hana2b
    SnapMirror Policy Name: SV_HANA
              Policy Owner: vserver-admin
               Tries Limit: 8
         Transfer Priority: normal
 Ignore accesstime Enabled: false
   Transfer Restartability: always
                   Comment: -
     Total Number of Rules: 2
                Total Keep: 8
                     Rules: Snapmirror-label  Keep Preserve Warn
                            ---------------- ---- -------- ----
                            daily              20  false      0
                            hourly             10  false      0
```

The policy must contain rules for all retention classes (labels) that are used in the Snap Creator configuration. The above commands show how to create a dedicated SnapMirror policy SV_HANA

2. To create and start the SnapVault relationship on the cluster console of the backup cluster, run the following commands.

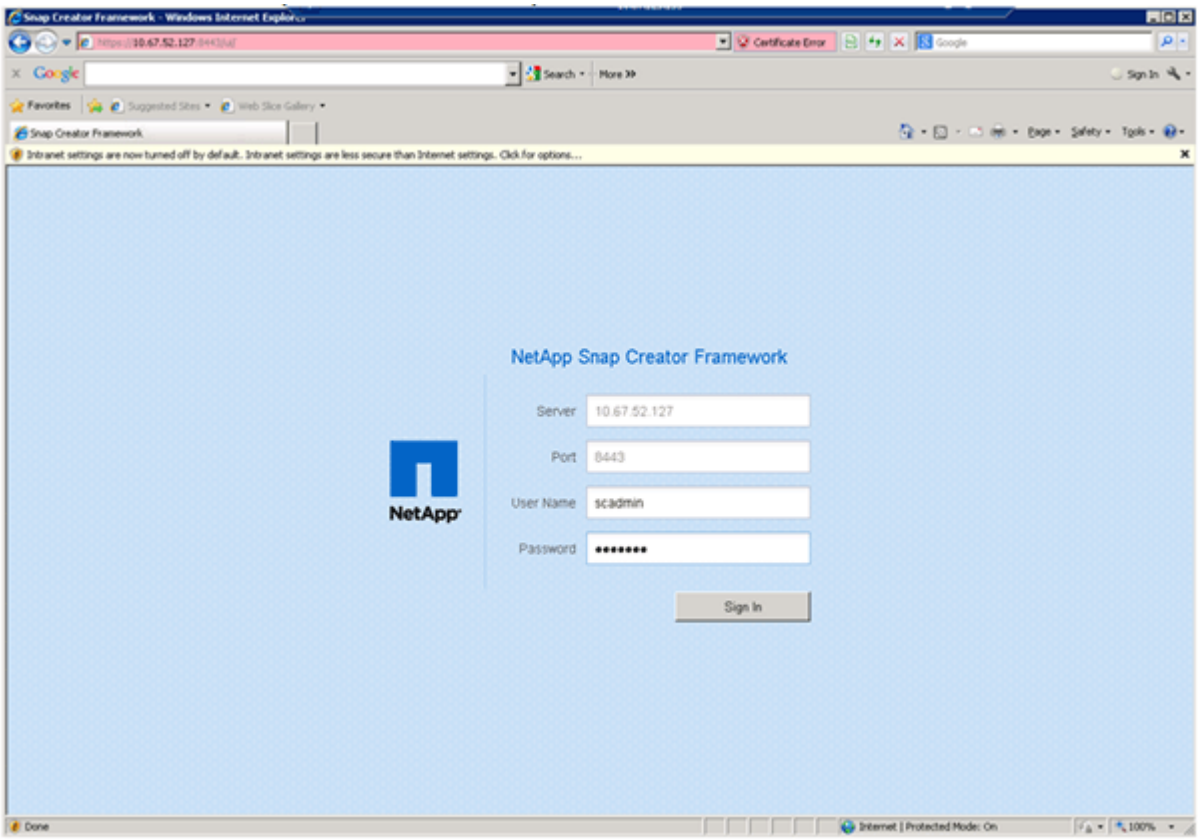
```
hana::> snapmirror create -source-path hana1a:hana_data -destination
-path
hana2b:backup_hana_data -type XDP –policy SV_HANA
Operation succeeded: snapmirror create the relationship with destination
hana2b:backup_hana_data.

hana::> snapmirror initialize -destination-path hana2b:backup_hana_data
-type XDP
```

Configuring the Snap Creator Framework and SAP HANA database backup

You must configure the Snap Creator Framework and the SAP HANA database backup.

1. Connect to the Snap Creator graphical user interface (GUI): https://host:8443/ui/.

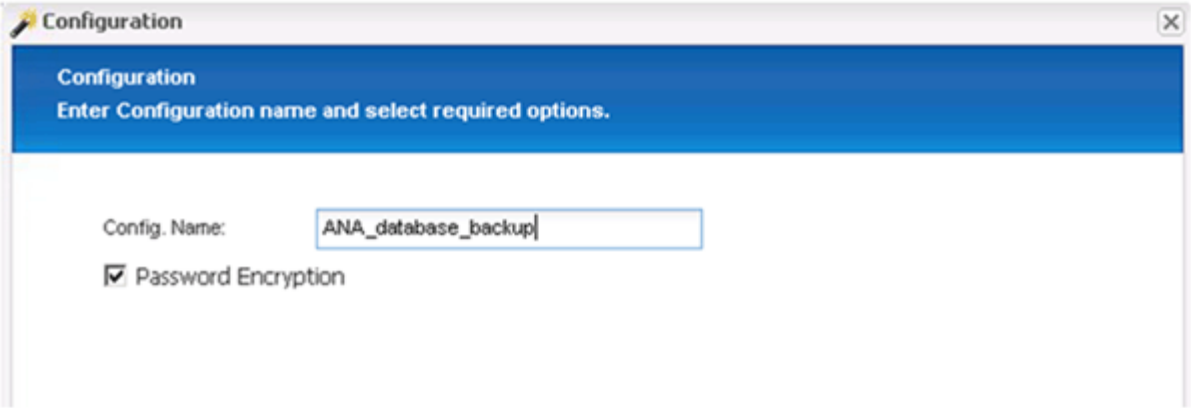
2. Log in using the user name and password that were configured during the installation. Click **Sign in**.



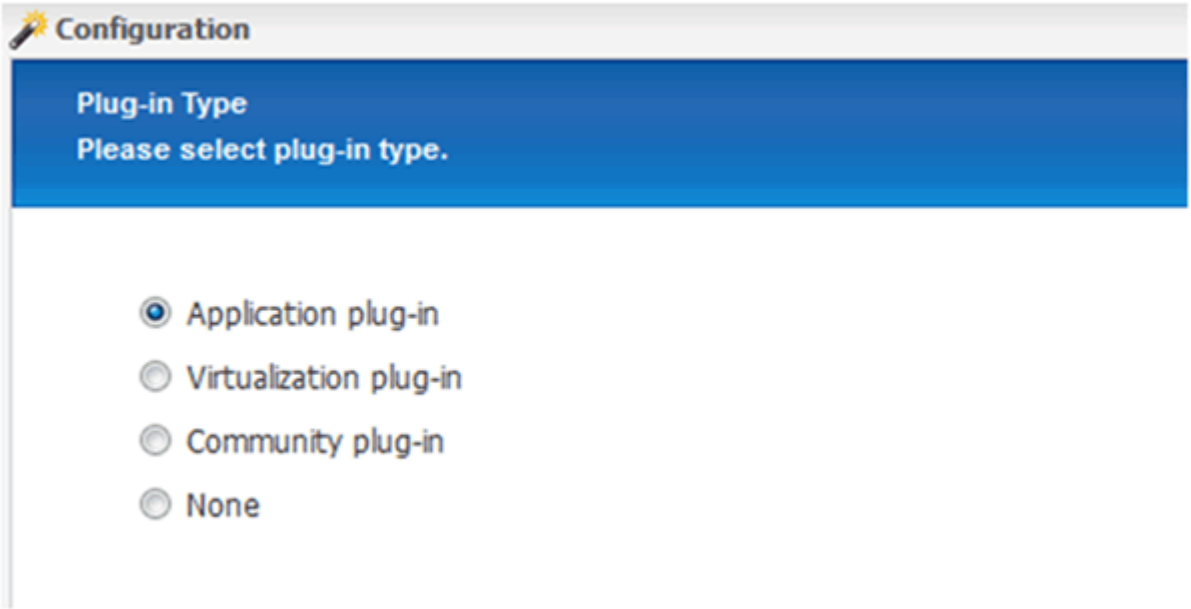
3. Enter a profile name and click **OK**.

For example, “ANA” is the SID of the database.

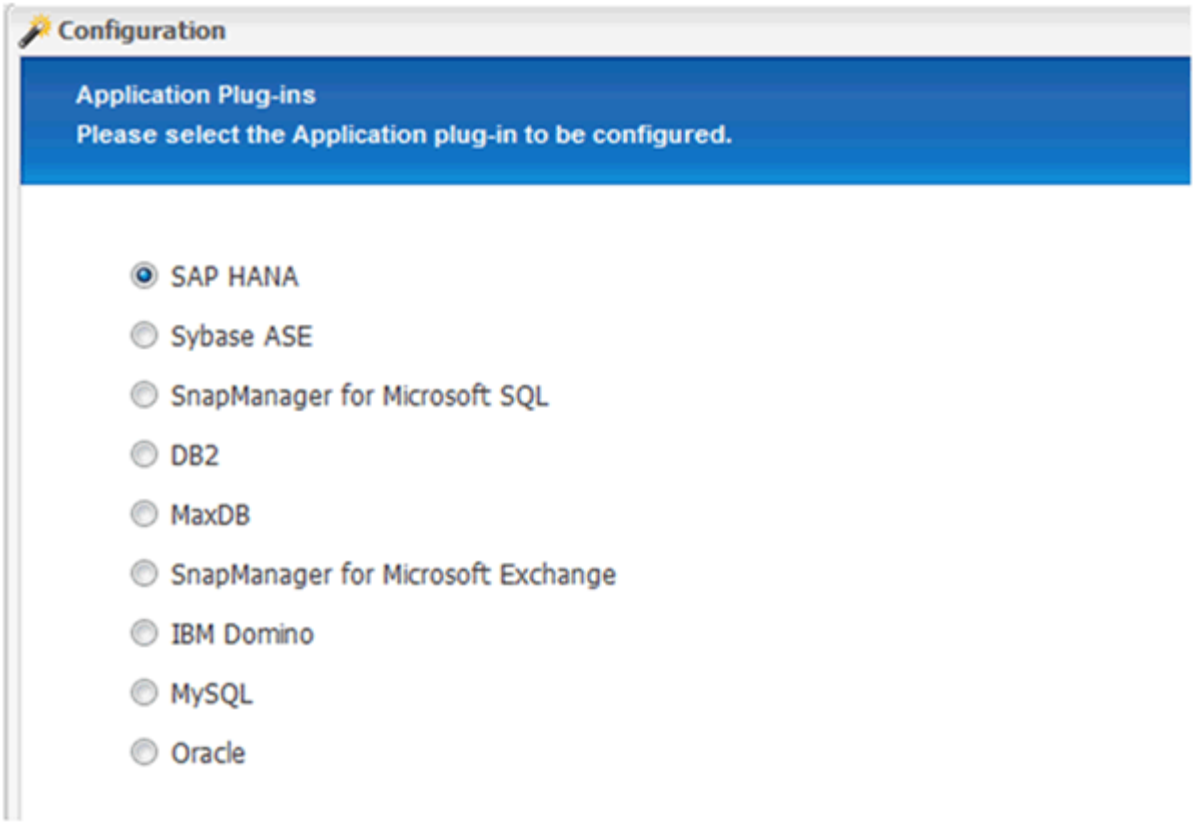
4. Enter the configuration name, and click **Next**.



5. Select **Application plug-in** as the plug-in type, and click **Next**.



6. Select **SAP HANA** as the application plug-in, and click **Next**.



7. Enter the following configuration details:

   a. Select **Yes** from the drop-down menu to use the configuration with a multitenant database. For a single container database select **No**.

   b. If Multitenant Database Container is set to **No**, you must provide the database SID.

   c. If Multitenant Database Container is set to **Yes**, you must add the hdbuserstore keys for each SAP HANA node.

   d. Add the name of the tenant database.

   e. Add the HANA nodes on which the hdbsql statement must be executed.

   f. Enter the HANA node instance number.

   g. Provide the path to the hdbsql executable file.

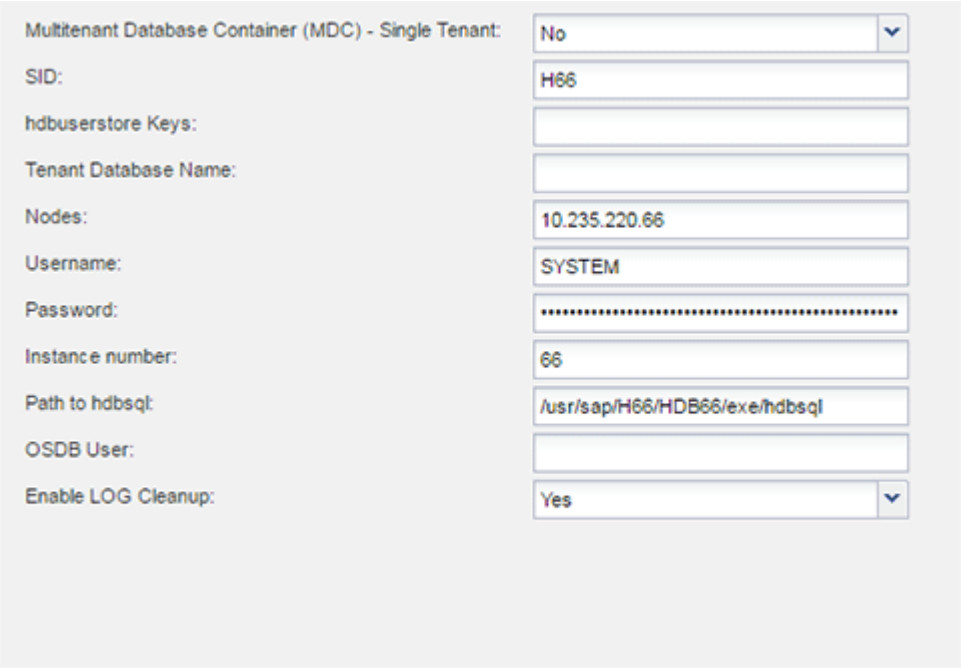
   h. Add the OSDB user.

   i. Select **Yes** from the drop-down list to Enable LOG Cleanup.

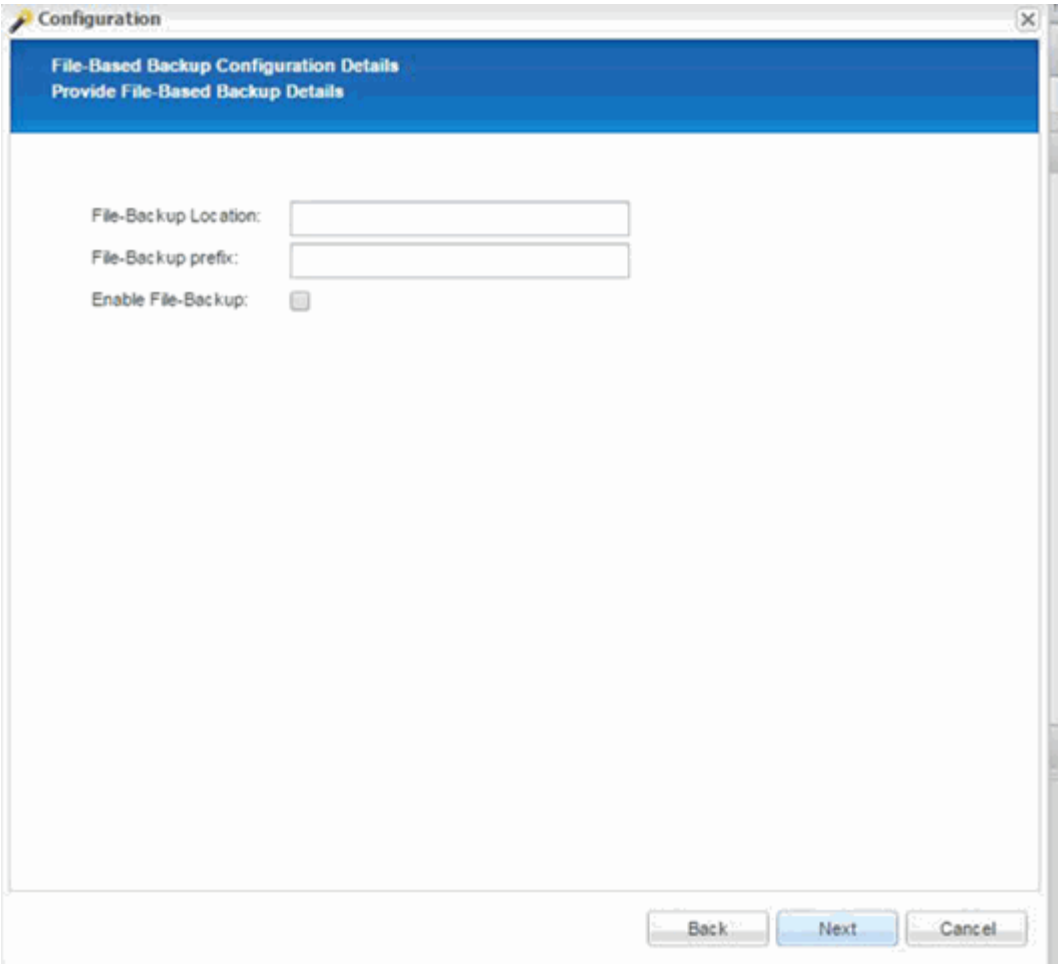
   NOTE:

   - Parameter `HANA_SID` is available only if the value for parameter `HANA_MULTITENANT_DATABASE` is set to `N`
   - For multitenant database containers (MDC) with a “Single Tenant” resource type, the SAP HANA Snapshot copies work with UserStore Key based authentication. If the `HANA_MULTITENANT_DATABASE` parameter is set to `Y`, then the `HANA_USERSTORE_KEYS` parameter must be set to the appropriate value.
   - Similar to non-multitenant database containers, the file-based backup and integrity check feature is supported

j. Click **Next**.



8. Enable the File-Based Backup operation:
   a. Set the File-Backup Location.
   b. Specify the file-backup prefix.
   c. Select the **Enable File-Backup** checkbox.
   d. Click **Next**.

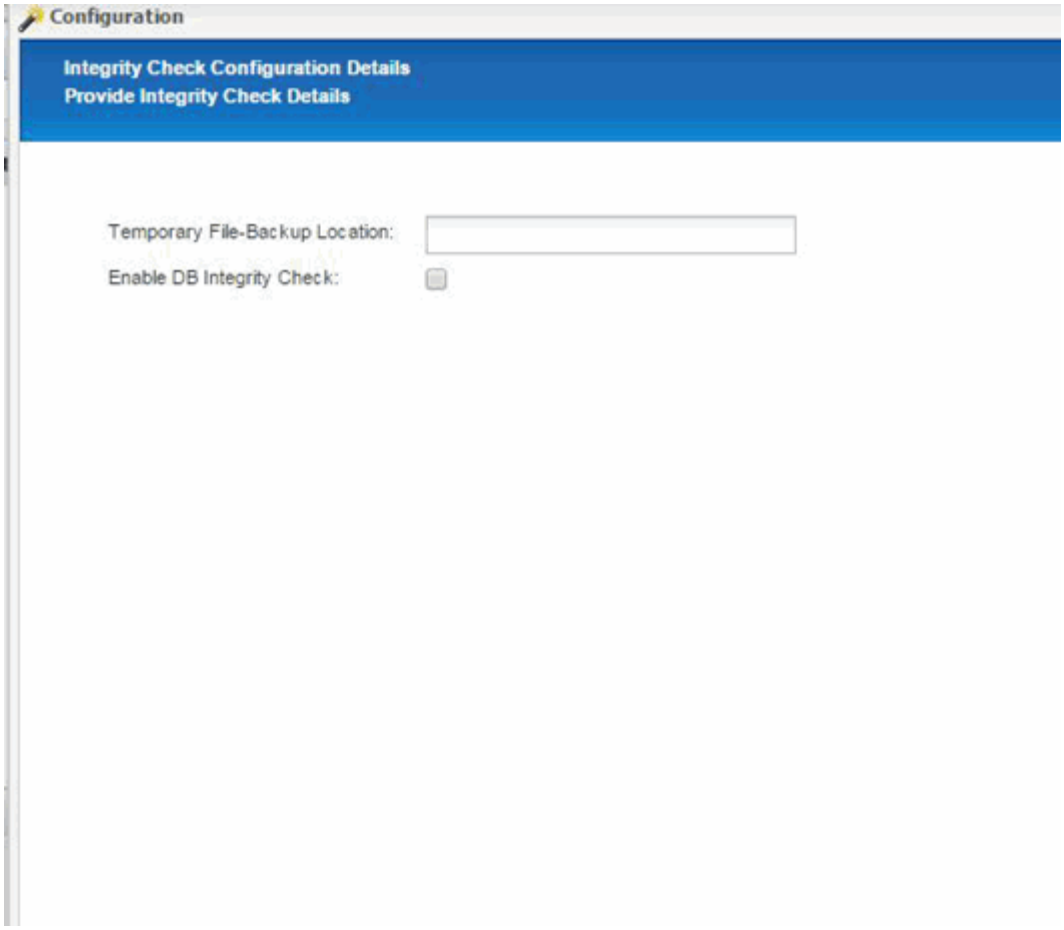


9. Enable the Database Integrity Check operation:

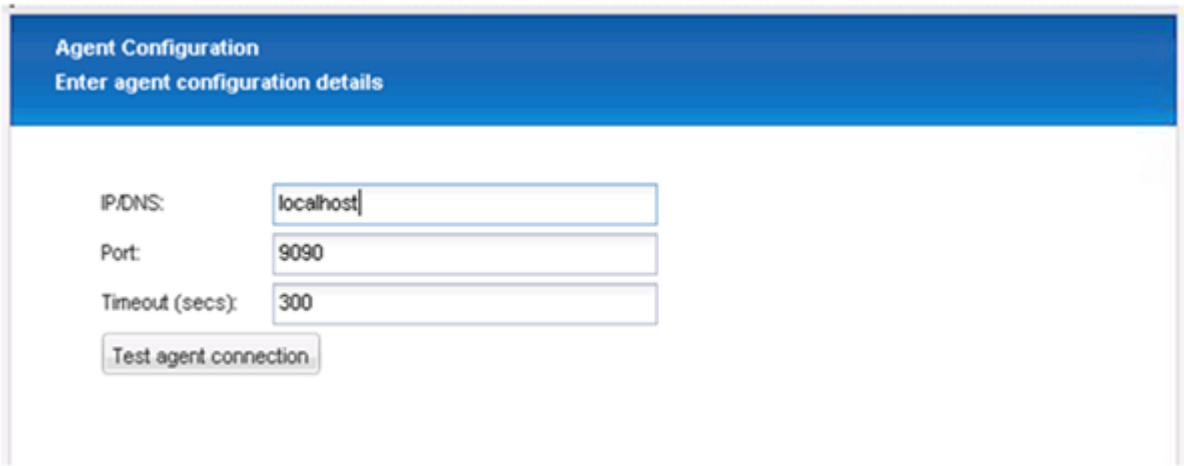
    a. Set the temporary File-Backup location.

    b. Select the **Enable DB Integrity Check** checkbox.

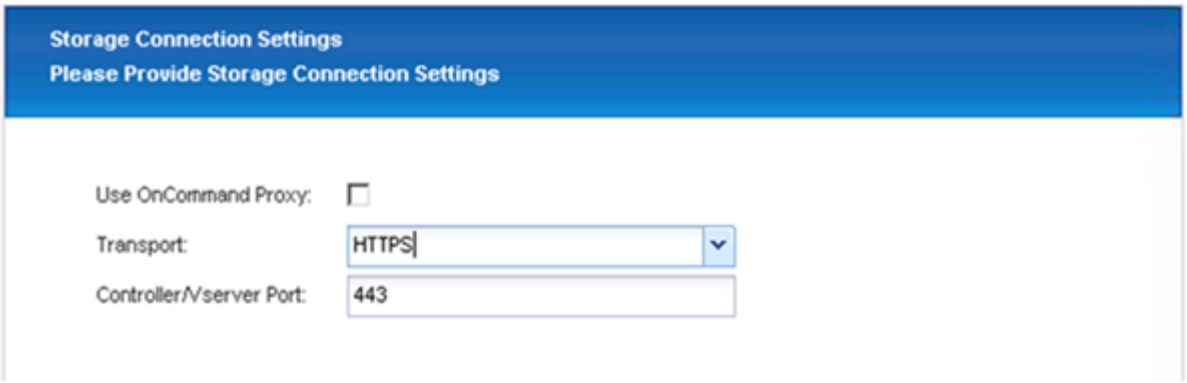
    c. Click **Next**.



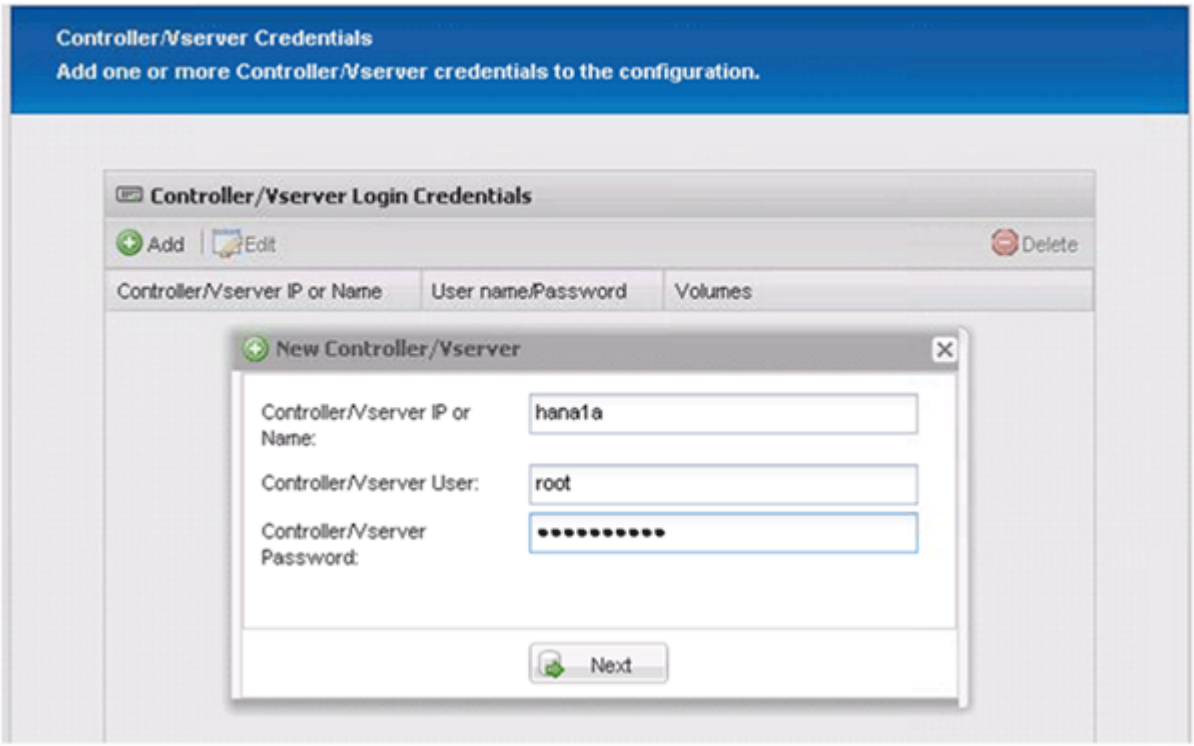
10. Enter the details for the agent configuration parameter, and click **Next**.



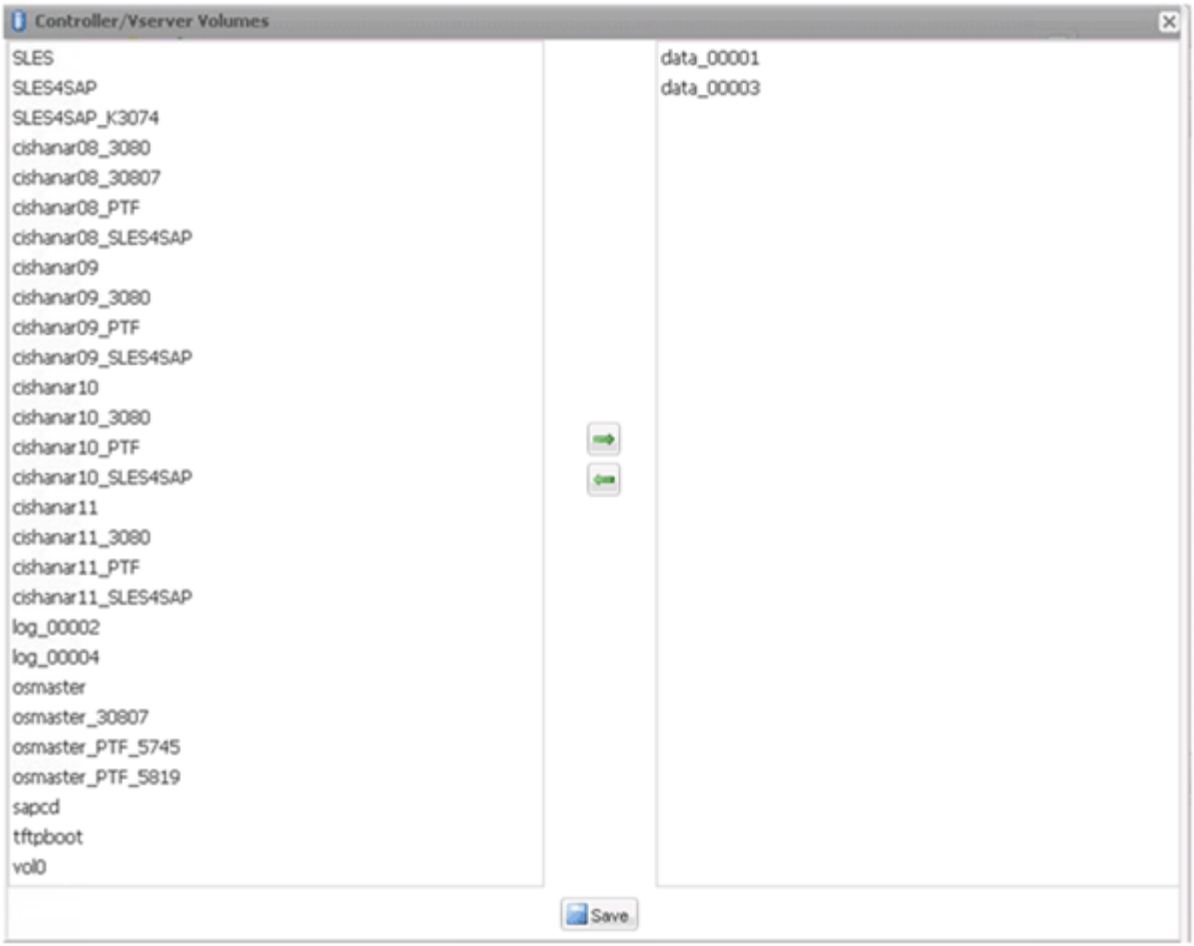
11. Enter the storage connection settings, and click **Next**.



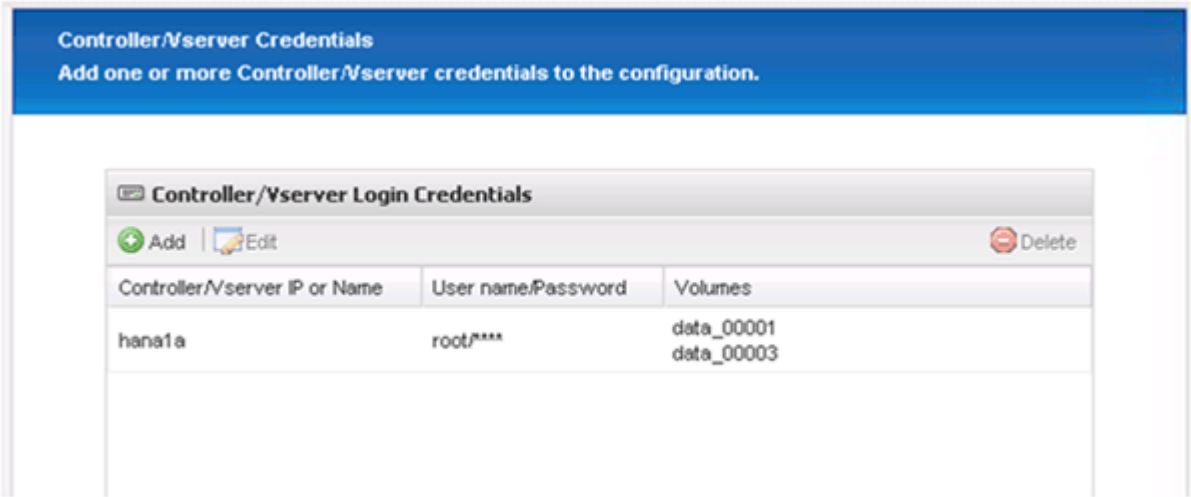
12. Enter the storage login credentials, and click **Next**.



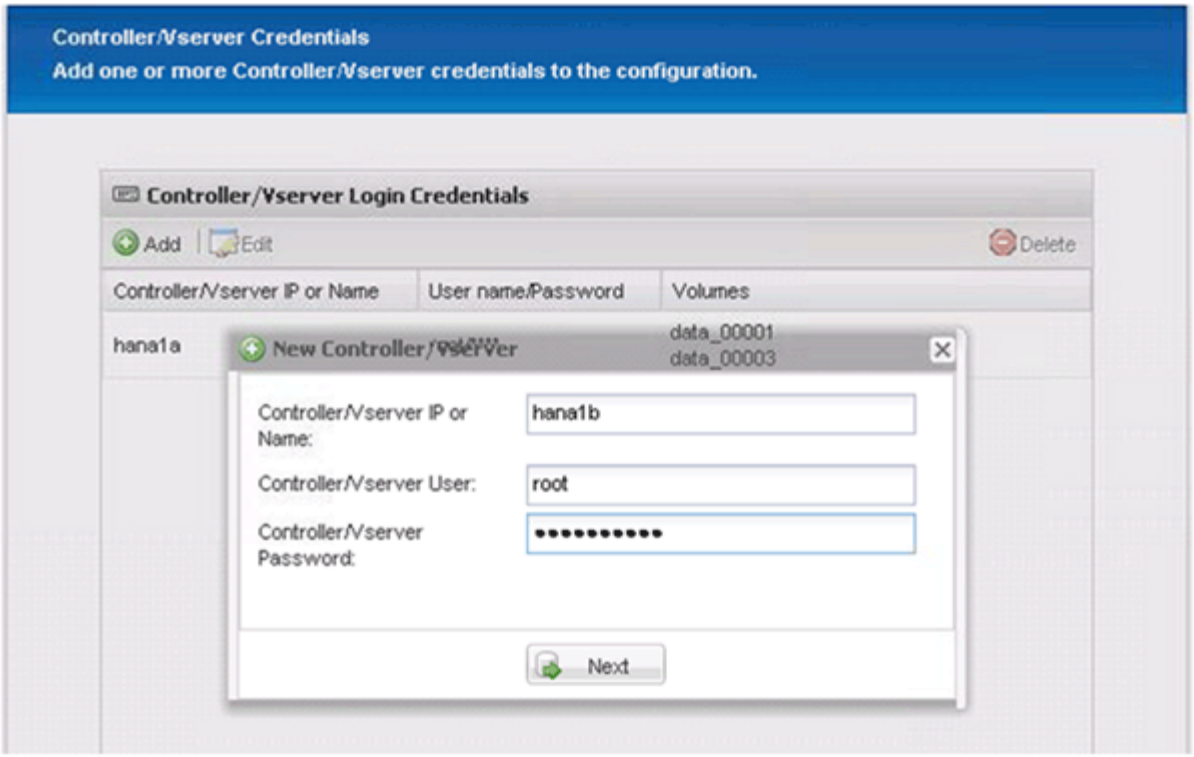
13. Select the data volumes that are stored on this storage controller, and click **Save**.



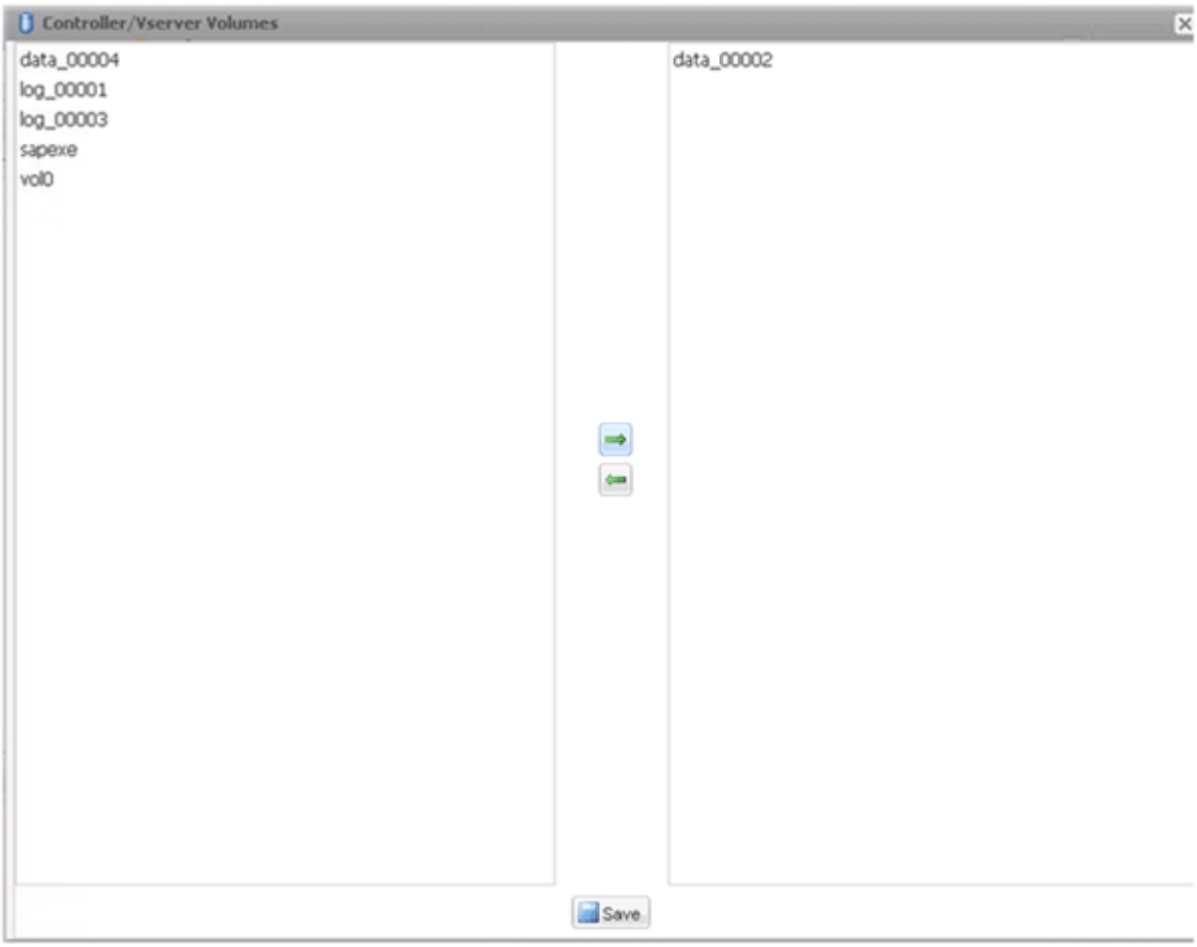
14. Click **Add** to add another storage controller.



15. Enter the storage login credentials, and click **Next**.

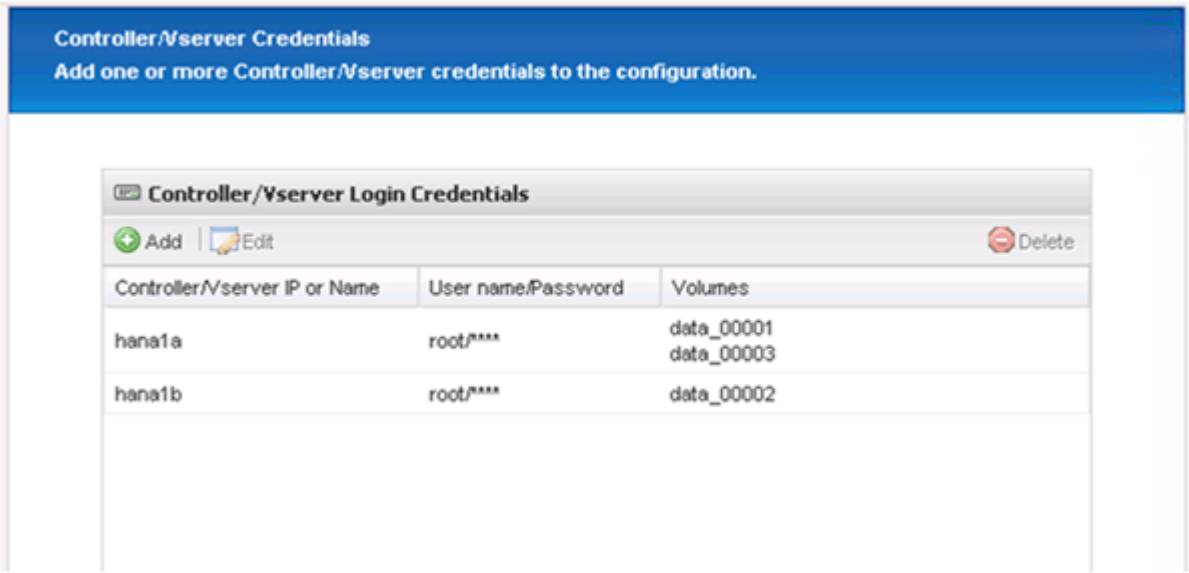


16. Select the data volumes that are stored on the second storage controller that you created, and click **Save**.



17. The Controller/Vserver Credentials window displays the storage controllers and volumes that you added.

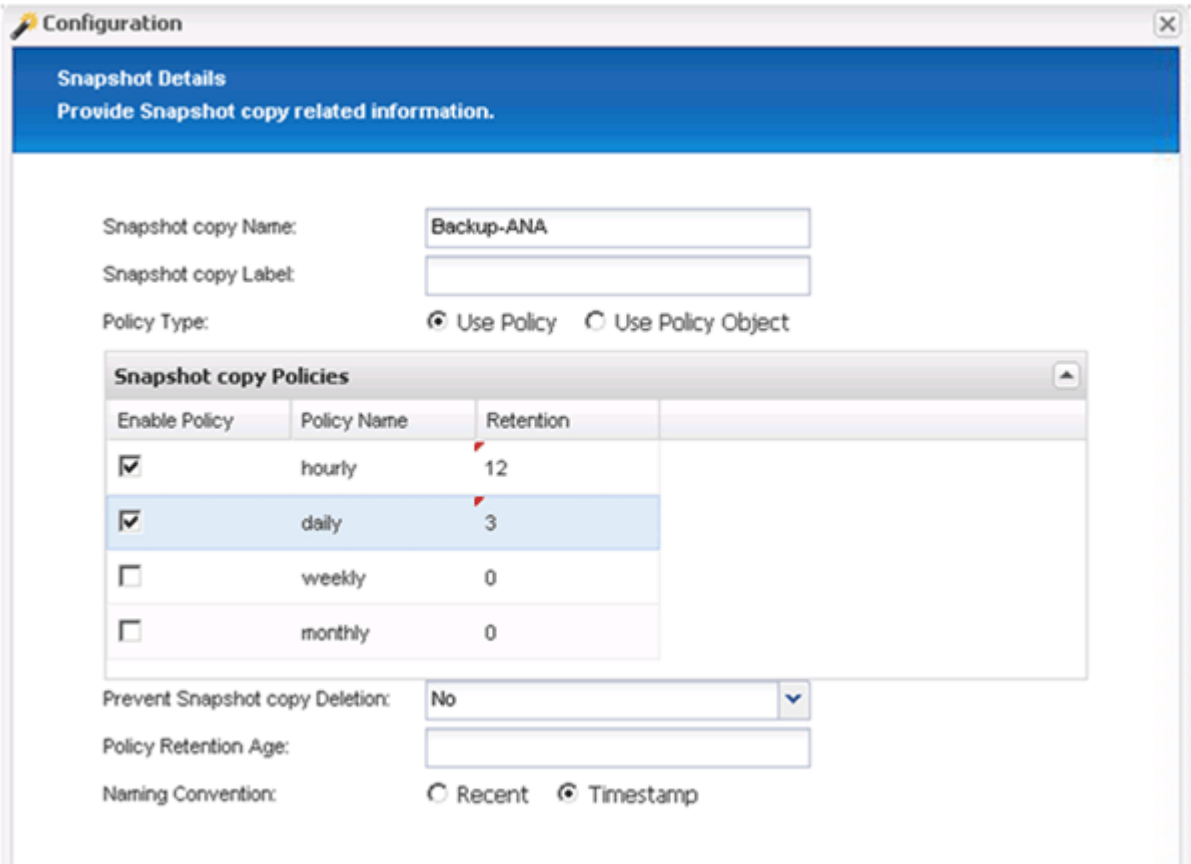
Click **Next**.



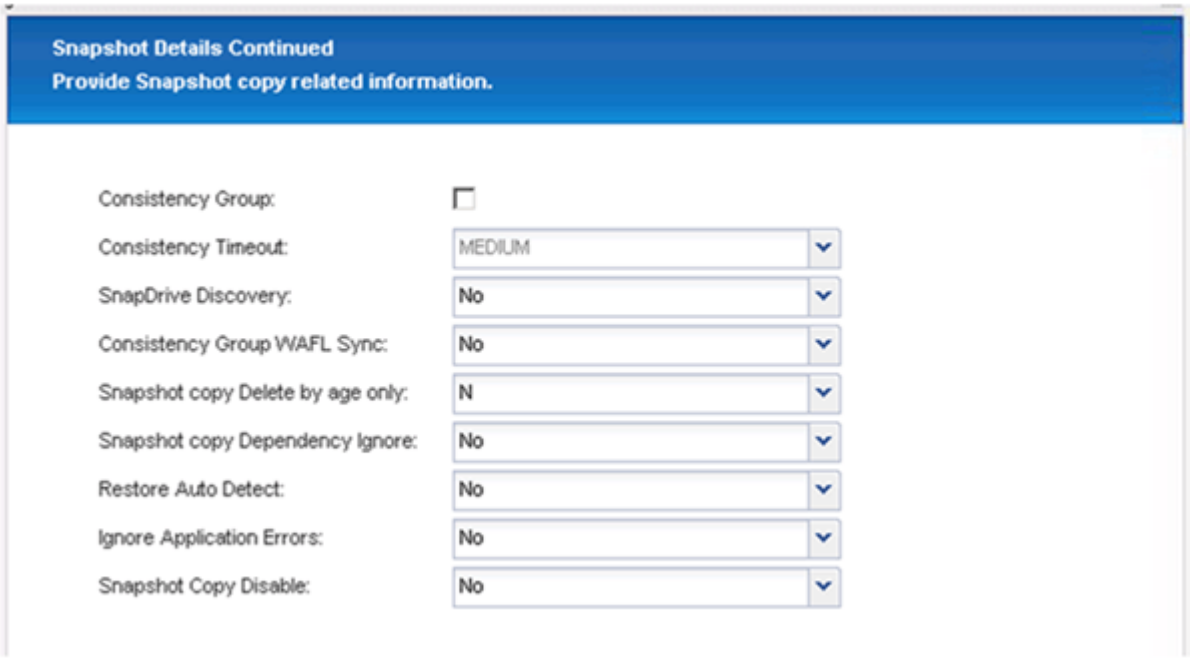
18. Enter the Snapshot policy and retention configuration.

    The retention of three daily and eight hourly Snapshot copies is just an example and could be configured differently depending on the customer requirements.

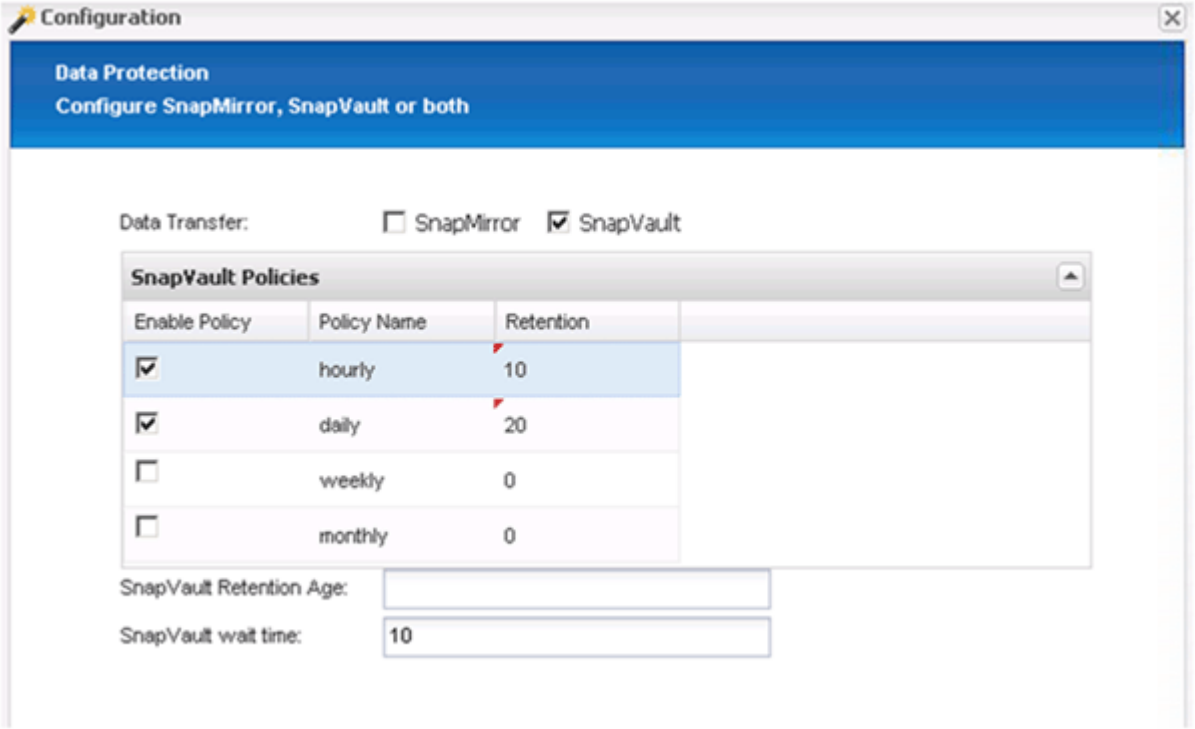
> Select **Timestamp** as the naming convention. The use of the naming convention **Recent** is not supported with the SAP HANA plug-in, because the timestamp of the Snapshot copy is also used for the SAP HANA backup catalog entries.



19. No changes required. Click **Next**.

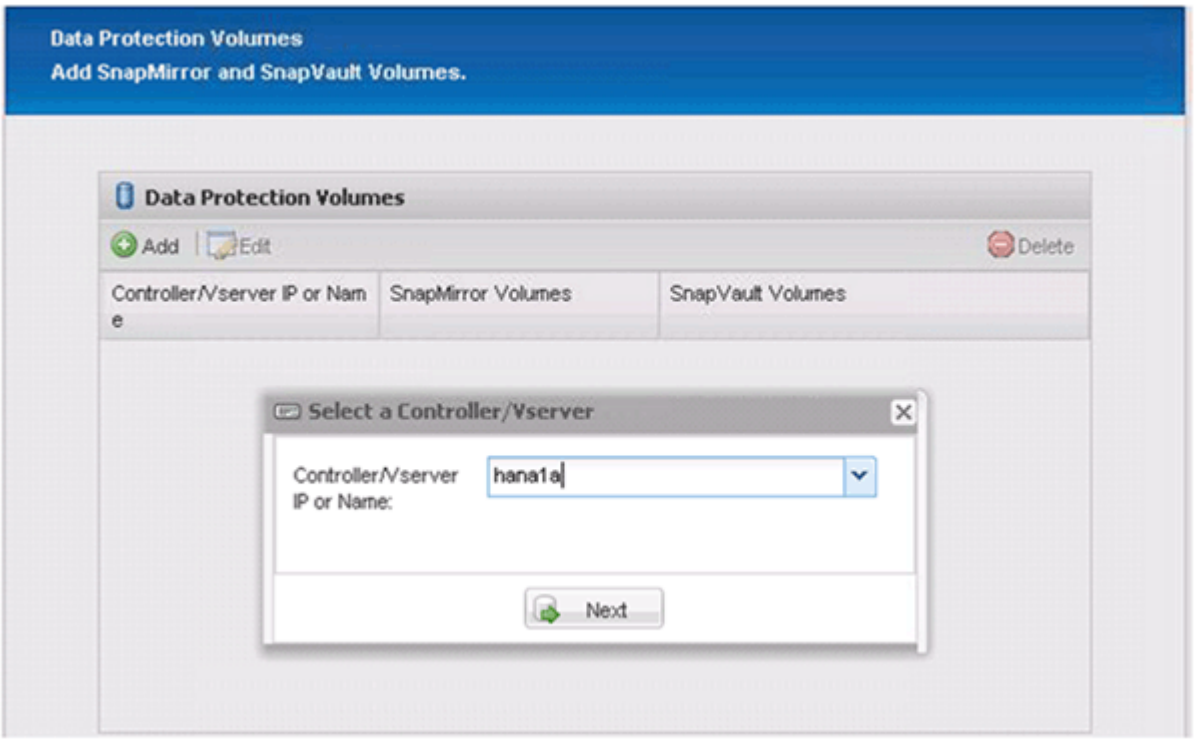


20. Select **SnapVault**, and configure the SnapVault retention policies and the SnapVault wait time.

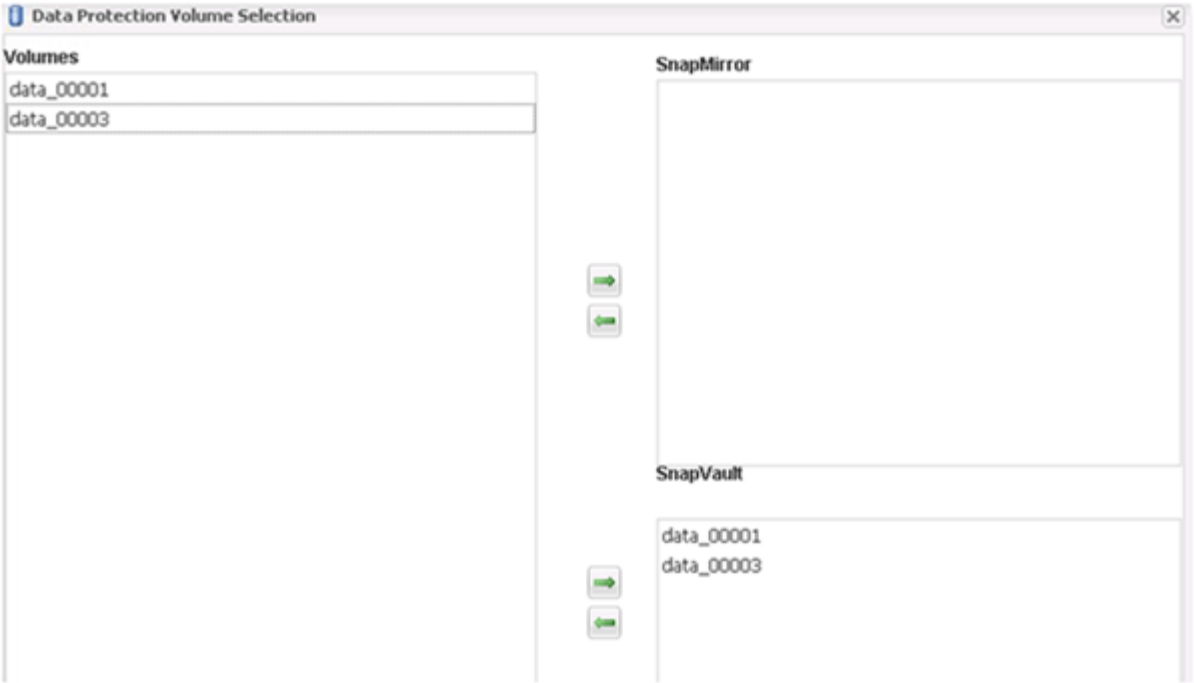


21. Click **Add**.

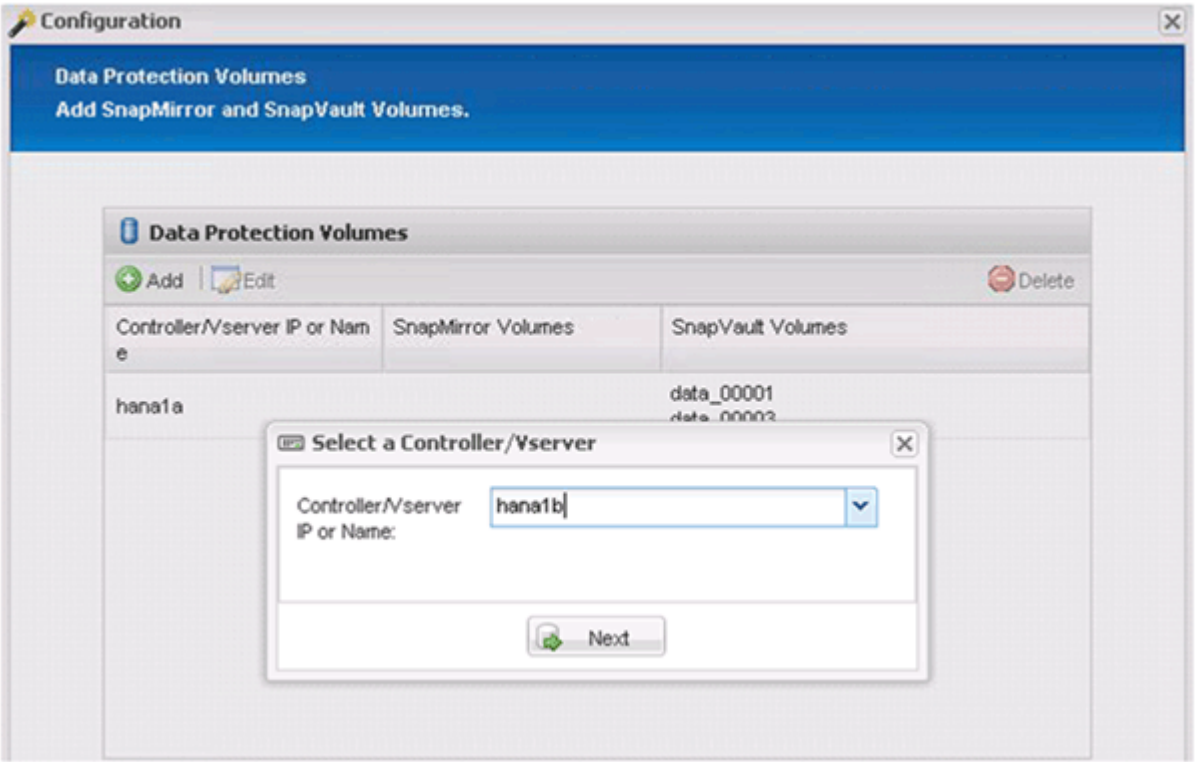
22. Select a source storage controller from the list, and click **Next**.



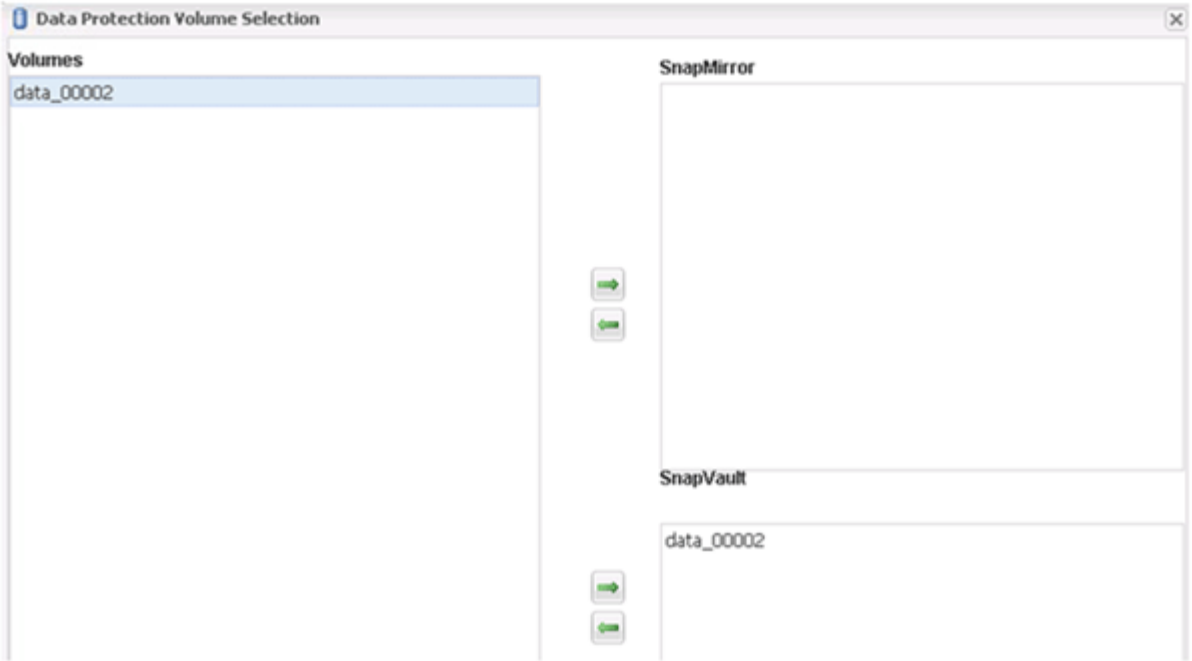
23. Select all the volumes that are stored on the source storage controller, and click **Save**.



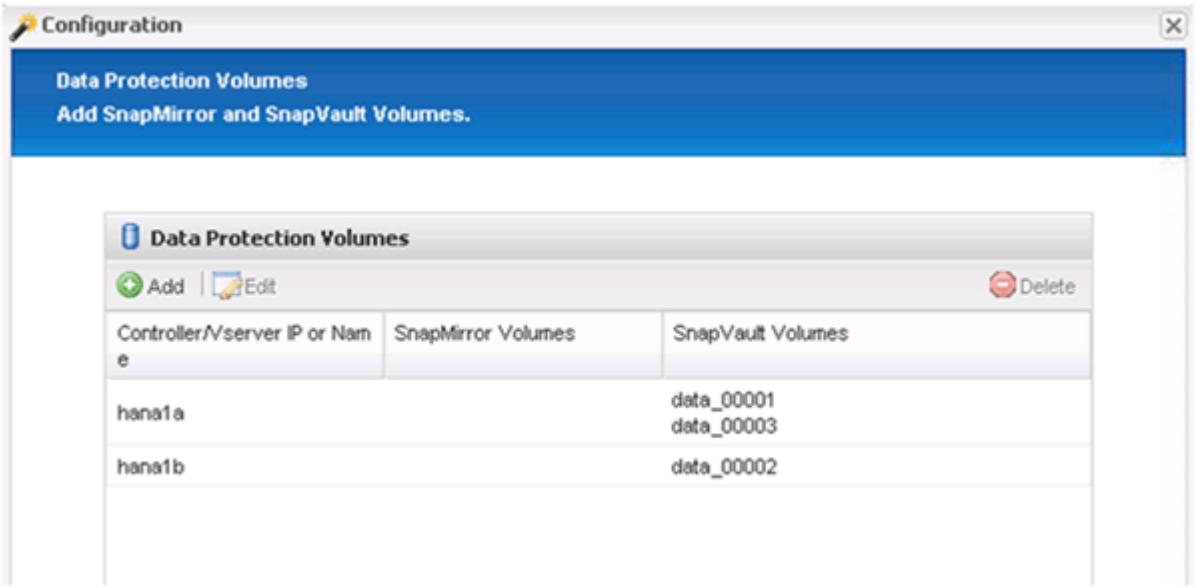
24. Click **Add**, and select the second source storage controller from the list, and then click **Next**.



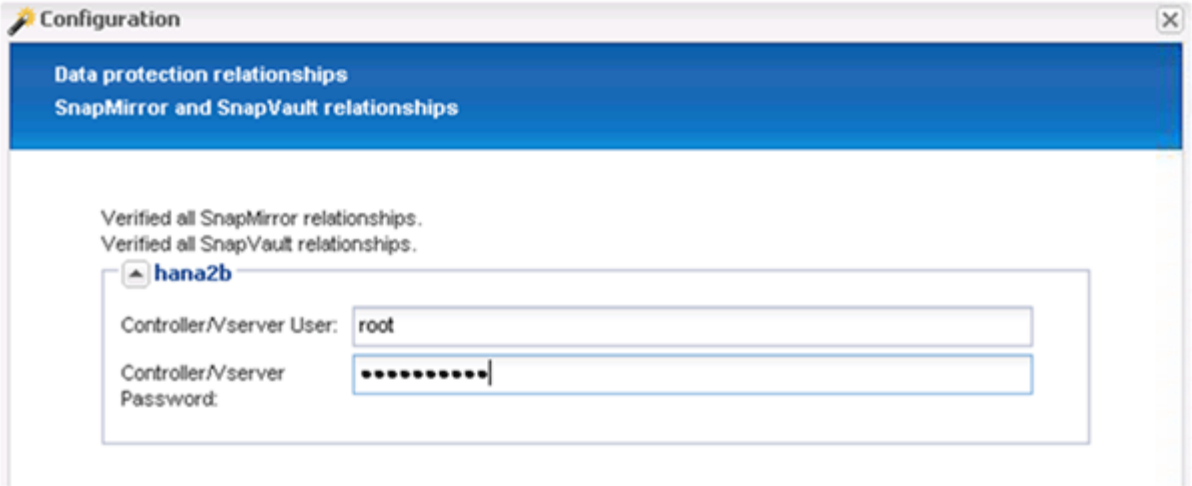
25. Select all the volumes that are stored on the second source storage controller, and click **Save**.



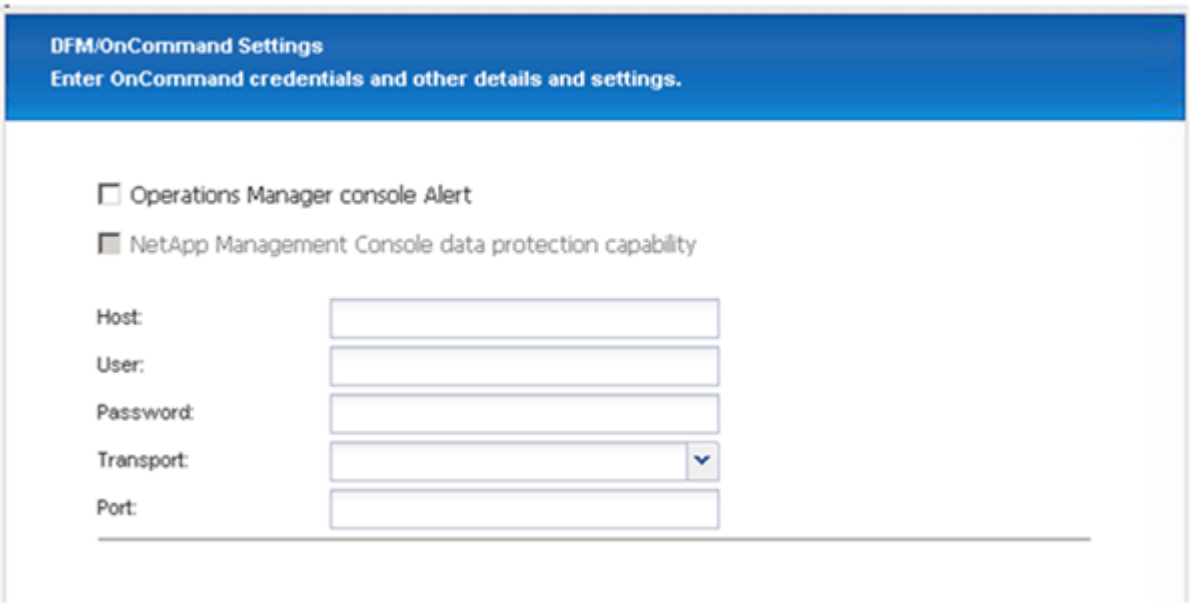
26. The Data Protection Volumes window displays all the volumes that should be protected in the configuration that you created. Click **Next**.



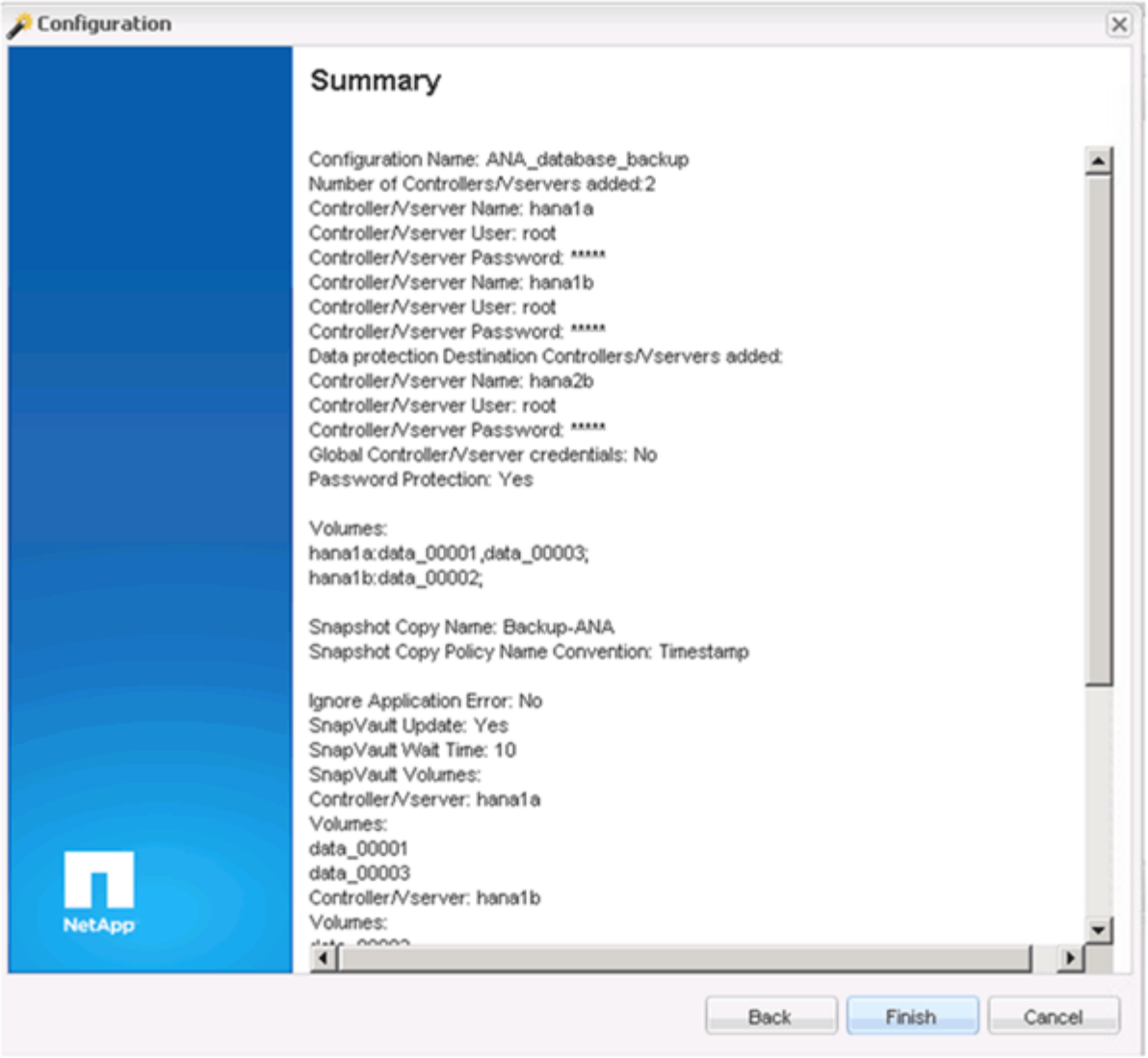
27. Enter the credentials for the target storage controllers, and click **Next**. In this example, the “root” user credentials are used to access the storage system. Typically, a dedicated backup user is configured on the storage system and is then used with Snap Creator.



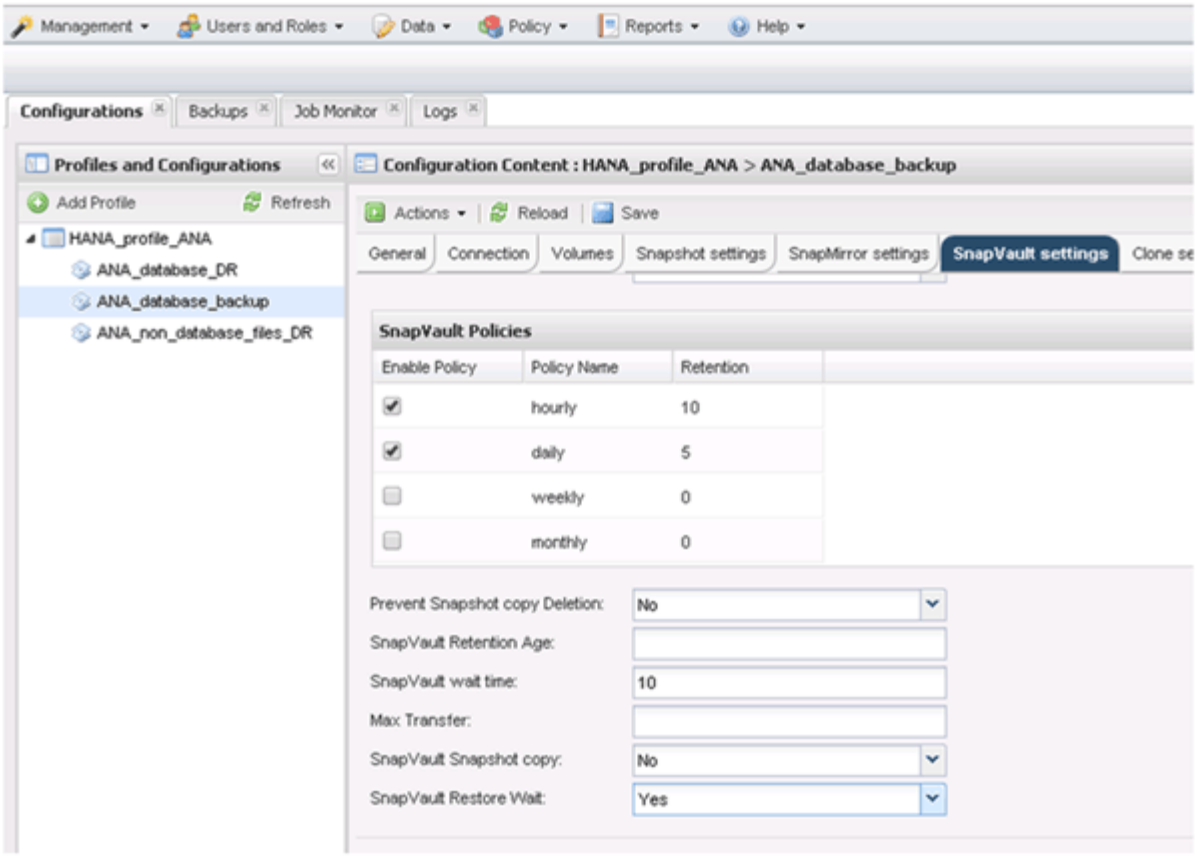
28. Click **Next**.



29. Click **Finish** to complete the configuration.



30. Click the **SnapVault settings** tab.
31. Select **Yes** from the drop-down list of the **SnapVault Restore Wait** option, and click **Save**.



It is recommended that you use a dedicated network for replication traffic. If you decide to do so, you should include this interface in the Snap Creator configuration file as a secondary interface.

You can also configure dedicated management interfaces so that Snap Creator can access the source or the target storage system by using a network interface that is not bound to the storage controller's host name.

```
mgmtsrv01:/opt/NetApp/Snap_Creator_Framework_411/scServer4.1.1c/engine/c
onfigs/HANA_profile_ANA
# vi ANA_database_backup.conf

#################################################################
#######################
#     Connection Options                                        #
#################################################################
#######################
PORT=443
SECONDARY_INTERFACES=hana1a:hana1a-rep/hana2b;hana1b:hana1b-rep/hana2b
MANAGEMENT_INTERFACES=hana2b:hana2b-mgmt
```

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